

Akshat Kankani

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EDUCATION

Manipal Institute of Technology

Bachelor of Technology in Computer Science (Artificial Intelligence)

- CGPA: 8.11

Bengaluru, KA

July 2023 – May 2027

PROJECTS

BiasScope | *Python, Hugging Face Transformers, PostgreSQL, FastAPI, Next.js*

Live Demo | GitHub

- Architected a distributed news intelligence platform using FastAPI, PostgreSQL, Redis, and Celery that transforms large-scale multi-source media streams into structured claims, evidence graphs, and event-level intelligence.
- Designed a claim-centric data model and retrieval architecture linking extracted claims, supporting evidence, publishers, and events, enabling efficient semantic retrieval, cross-source corroboration, and evidence-grounded analysis.
- Built scalable asynchronous processing pipelines for ingestion, enrichment, clustering, and intelligence generation, reducing recomputation through incremental snapshot-based processing of newly published content.
- Developed transformer-based NLP services for bias analysis, factual claim extraction, evidence attribution, and semantic clustering, enabling automatic discovery of corroborated events across independent publishers.
- Engineered auditable intelligence workflows including source-reliability scoring, narrative anomaly detection, diversity metrics, and longitudinal narrative tracking to improve transparency, trustworthiness, and reproducibility of generated reports.

DataScope | *Python, Pandas, scikit-learn, PostgreSQL, FastAPI*

Live Demo | GitHub

- Architected a machine learning governance platform that evaluates dataset quality, feature integrity, model reliability, and deployment readiness through automated validation, leakage detection, anomaly analysis, and governance scoring.
- Built a comprehensive validation and explainability engine combining statistical profiling, feature leakage detection, SHAP attribution, permutation importance, feature ablation, and remediation impact simulation to identify deployment risks and improve model trustworthiness.
- Designed a consensus-based anomaly detection framework integrating Isolation Forest, DBSCAN, Z-score, and MAD analysis with identifier-aware preprocessing, reducing false positives and improving anomaly classification accuracy across heterogeneous datasets.
- Implemented an asynchronous analysis platform using FastAPI, Celery, Redis, and PostgreSQL to orchestrate dataset ingestion, governance evaluation, model training, explainability workflows, deployment recommendations, and end-to-end audit trail generation.

PRScope | *Python, PostgreSQL, FastAPI, Github OAuth, Typescript, React*

GitHub

- Built PRScope, a Chrome extension and FastAPI-based code review platform for GitHub pull requests, supporting automated risk assessment, security analysis, dependency impact evaluation, review tracking, and GitHub comment publishing.
- Designed a deterministic review engine combining PR metadata, changed symbols, AST-based code analysis, dependency graphs, call graph tracing, architecture rules, and test coverage signals to generate explainable risk scores and review recommendations.
- Developed static and security analysis pipelines to identify impacted code paths, dependency blast radius, SQL injection patterns, hardcoded secrets, unsafe deserialization, dynamic code execution, and command injection vulnerabilities.
- Integrated GitHub OAuth, PostgreSQL-backed review workspaces, Jira requirement alignment, historical review timelines, and a Gemini-powered review assistant for PR classification, diff summarization, confidence-ranked comments, and executive review summaries.

TECHNICAL SKILLS

Languages: Python, C, C++

AI & ML: PyTorch, scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib, Seaborn

Databases: MongoDB, PostgreSQL

Developer Tools & Frameworks: FastAPI, Git, GitHub, Docker, Kubernetes, Redis, Celery, Postman, PowerBI

Coursework: DSA, Machine Learning, Neural Networks & Deep Learning (Fundamentals), OOP, CN, DBMS